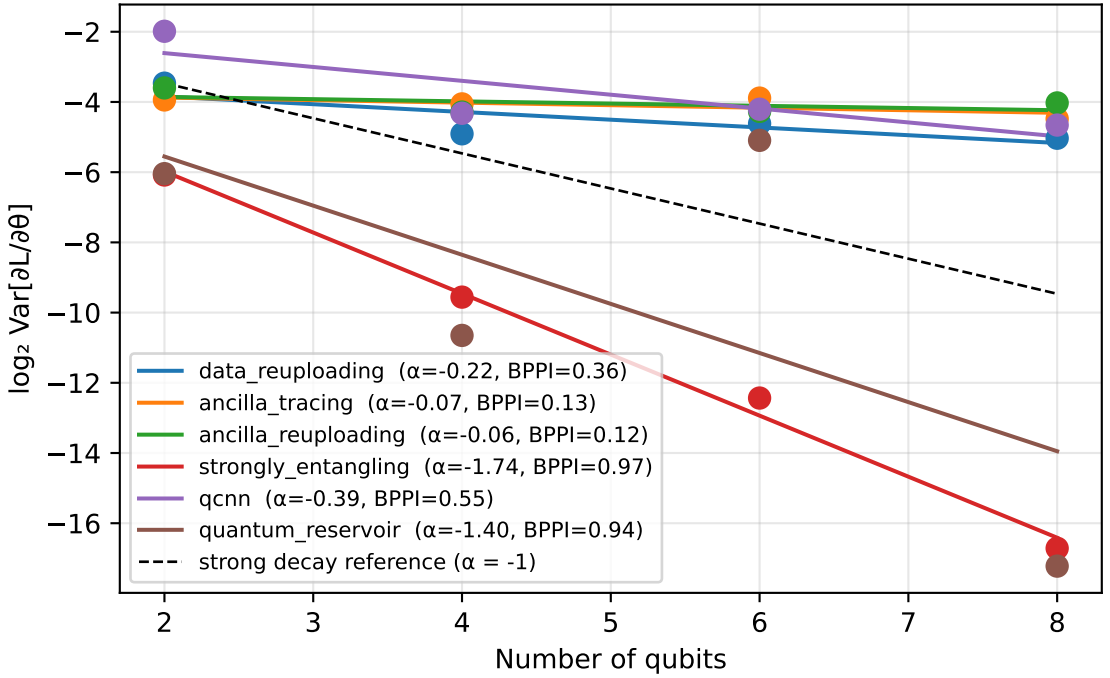
Gradient variance scaling — barren plateau analysis
log₂ Var[∂L/∂θ]
−2
−4
−6
−8
−10
−12
−14
−16
2
3
4
5
6
7
8
Number of qubits
data_reuploading (α=-0.22, BPPI=0.36)
ancilla_tracing (α=-0.07, BPPI=0.13)
ancilla_reuploading (α=-0.06, BPPI=0.12)
strongly_entangling (α=-1.74, BPPI=0.97)
qcnn (α=-0.39, BPPI=0.55)
quantum_reservoir (α=-1.40, BPPI=0.94)
strong decay reference (α = -1)